Supplementary Material for "Adaptive calibration-aware pooling for threshold-exceedance risk assessment and early warning in deep excavations"

Abstract

This Supplementary Material provides the implementation, methodological, and diagnostic detail supporting the main paper. It describes the construction and operational interpretation of the pooling layer, including the roles of the two pooling families, the common spread adjustment, and tail completion, and gives the full specifications of the five probabilistic forecasting members together with their common probabilistic output. It further sets out the census of probability-integral-transform flags and the per-series modelling roles, details the selection of the pre-deployment prior, provides stratified calibration and dependence diagnostics, and presents the event-level warning audit for the independent transfer project with per-series bookkeeping and a rate-episode audit. It then collects the ablation and sensitivity analyses of the member set, the selection rule, the online updates, the buffer length, the anchoring strength, the coverage floor, the spread factors, the tail completion, the robust scale, the choice of global sequence member, and the optimiser seeds, adds online conformal recalibration as a reference, gives crossed-group, stage-wise, and member-level diagnostics, and summarises the ground conditions of the two projects.

S1. Pooling-family construction

The adaptive calibration-aware pooling (ACP) framework uses the two alternative pooling families defined in Eqs. (2)–(3) of the main text, quantile averaging (QA) and the spread-adjusted linear pool (SLP), the latter a linear pool in which the cumulative distribution function (CDF) of each member is first rescaled about its own median. Both families combine the same five member predictive distributions using a common weight vector $\mathbf{w}$ and a spread factor τ. The two families differ only in the representation being combined, namely member quantiles for QA and member CDFs for SLP.

In the SLP family, each member CDF is evaluated on the ordered union of the quantile knots of all five members, whatever their weights, and the pooled CDF is interpolated linearly between consecutive knots. Below the rescaled 0.01 quantile of a member its CDF is taken as zero and above the rescaled 0.99 quantile as one, so the one per cent tail mass on each side of a member is spread uniformly between its outer stored level and the adjacent outer knot of the union, or placed at that level when it is the outermost knot. The interpolant therefore equals the pointwise mixture of Eq. (3) of the main text except on the union intervals that end at the rescaled 0.01 quantile or start at the rescaled 0.99 quantile of a member. No member quantile is repeated in the stored forecasts of either project, so no member CDF has a vertical segment within its stored range. The pooled CDF is monotone by construction, and a running

maximum is applied only as a numerical safeguard against floating-point non-monotonicity before the pooled CDF is inverted at the levels of the common grid. A census of the issued distributions found no repeated pooled quantile on either project.

After pooling, the interior distribution is completed with the common tail-completion rule described in the main text. The completed distribution is monotone over the full response range, and quantiles, exceedance probabilities, and warning decisions are derived from it. The same rule is applied to all candidate pools so that their scores are comparable. The completed distributions are used for the continuous ranked probability score (CRPS) of every candidate pool and for the member losses of the online layer. By contrast, the probability integral transform (PIT) values of the calibration objective, of the PIT Cramér–von Mises (CvM) discrepancies, and of the PIT histograms use the piecewise-linear interior CDF, and an outcome below the 0.01 or above the 0.99 stored level receives the PIT value 0.005 or 0.995, the midpoint of the one per cent of probability beyond that level. The PIT flags and the cross-instrument dependence estimates use the PIT of the completed distribution.

The common spread factor τ adjusts the dispersion of the pooled distribution and is chosen with the weights in the pre-deployment pool selection. In the QA family, it rescales quantile deviations from the pooled median without changing the median itself. In the SLP family, it rescales each member CDF about the median of that member.

S2. Probabilistic member specifications

The following specifications complete the implementation details summarised in Table 2 of the main text. The first three members are local, fitted separately to each monitoring series, whereas the Light Gradient Boosting Machine (LightGBM) member and the gated recurrent unit (GRU) ensemble are global, trained across the Project-A source panel. All origin-specific preprocessing, fitting, and forecast construction use only information available at or before the forecast origin. The training rows of the two global members are built from the data used for their fit. Within the member derivations, the series index is suppressed and x_t denotes the recorded value at t, which equals $x_j(t)$ except for the sign of a settlement series, consistent with Table 2 of the main text. As in the main text, the members are applied to the recorded series, so a settlement series, recorded negative downward, enters them without the change of sign.

For series j at origin t, the robust scale used by the standardised members is $s_{j,t} = 1.4826\,\mathrm{MAD}(\Delta x)$, the scaled median absolute deviation (MAD) computed from the increments Δx of the series available at or before the origin and bounded below by 0.05 mm. The exact fitting and architecture settings are collected in Table S1.

Local-linear-trend state-space model. The local-linear-trend (LLT) member uses the linear Gaussian state-space system specified in Table 2 of the main text. With $\mathbf{a}_t = (\mu_t, \nu_t)^\top$, the state evolution and observation equations can equivalently be expressed as $\mathbf{a}_{t+1} = \mathbf{T}\mathbf{a}_t + \boldsymbol{\omega}_t$ and $x_t = \mathbf{Z}\mathbf{a}_t + \varepsilon_t$, with $\mathbf{T} = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\mathbf{Z} = (1, 0)$. Here $\boldsymbol{\omega}_t = (\xi_t, \zeta_t)^\top$ collects the state disturbances of Table 2 of the main text. The corresponding process covariance is $\mathbf{Q} = \mathrm{diag}(\sigma_\xi^2, \sigma_\zeta^2)$ and the observation variance is σ_ε^2.

At each observed monitoring date, the Kalman prediction and update are performed as

$$\mathbf{a}_{t|t-1} = \mathbf{T}\mathbf{a}_{t-1|t-1}, \qquad \mathbf{P}_{t|t} = \mathbf{P}_{t|t-1} - \mathbf{K}_t\mathbf{Z}\mathbf{P}_{t|t-1},$$

where $\mathbf{P}_{t|t-1}$ and $\mathbf{P}_{t|t}$ denote the predicted and filtered state covariance matrices, with $\mathbf{P}_{t|t-1} = \mathbf{T}\mathbf{P}_{t-1|t-1}\mathbf{T}^\top + \mathbf{Q}$, and where the Kalman gain is $\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{Z}^\top/(\mathbf{Z}\mathbf{P}_{t|t-1}\mathbf{Z}^\top + \sigma_\varepsilon^2)$. The filtered state mean is updated as $\mathbf{a}_{t|t} = \mathbf{a}_{t|t-1} + \mathbf{K}_t(x_t - \mathbf{Z}\mathbf{a}_{t|t-1})$, where $x_t - \mathbf{Z}\mathbf{a}_{t|t-1}$ is the innovation. For missing monitoring dates, the state is propagated by the prediction step without an observation update, and no measurement is imputed.

The three variance parameters are estimated by Gaussian maximum likelihood (ML) from the series-specific initial history and are then kept fixed during the corresponding rolling evaluation. Repeated state propagation produces the h-step predictive distribution $N(\hat{x}_{t+h|t}, \sigma^2_{t+h|t})$.

Gaussian-process model. The Gaussian-process (GP) member uses a training window of up to the 60 most recent available observations, and within this member $s_{j,t}$ is computed from the increments inside that window. At each origin, responses are centred at the current level and scaled as $\tilde{x}_i = (x_i - x_t)/s_{j,t}$, where i indexes the observations in this training window. Elapsed monitoring time in days is the scalar input. The latent process uses $k_f(t_i, t_{i'}) = \sigma_f^2 \exp[-(t_i - t_{i'})^2/(2\ell_{\mathrm{GP}}^2)]$ and an independent Gaussian observation-noise variance σ_n^2.

For the vector $\tilde{\mathbf{x}}$ of the n training observations and the latent-kernel matrix $\mathbf{K}_f$, the log marginal likelihood is

$$-\frac{1}{2}\tilde{\mathbf{x}}^\top(\mathbf{K}_f + \sigma_n^2\mathbf{I})^{-1}\tilde{\mathbf{x}} - \frac{1}{2}\log|\mathbf{K}_f + \sigma_n^2\mathbf{I}| - \frac{n}{2}\log(2\pi).$$

Hyperparameters are estimated by marginal-likelihood maximisation, with ℓ_{GP} initialised at 10 d and the optimisation bounds listed in Table S1. In that table, GPR denotes the `scikit-learn` Gaussian-process regressor and Const, RBF, and White its constant, radial basis function, and white-noise kernel terms. For a future time t_*, let $\mathbf{k}_*$ denote the vector with entries $k_f(t_*, t_i)$ over the training times. The predictive mean and observed-response variance are

$$m_* = \mathbf{k}_*^\top(\mathbf{K}_f + \sigma_n^2\mathbf{I})^{-1}\tilde{\mathbf{x}}, \qquad v_* = k_f(t_*, t_*) + \sigma_n^2 - \mathbf{k}_*^\top(\mathbf{K}_f + \sigma_n^2\mathbf{I})^{-1}\mathbf{k}_*.$$

They are returned to the measurement scale as $\mu_* = x_t + s_{j,t}m_*$ and $\sigma_* = s_{j,t}\sqrt{v_*}$.

Rectangular-hyperbola model. This member fits $x(t) = \beta_0 + \beta_1 t/(\beta_2 + t)$ by bounded nonlinear least squares, where t is one day plus the elapsed time in days since the first observation of the series. The curve parameters are collected in $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^\top$. The fitted covariance $\widehat{\boldsymbol{\Sigma}}_\beta$ is obtained from the local least-squares curvature.

At forecast time t_*, parameter uncertainty is propagated by the delta method,

$$\sigma^2_{\mathrm{pred}}(t_*) = \mathbf{g}_*^\top\widehat{\boldsymbol{\Sigma}}_\beta\mathbf{g}_* + \hat{\sigma}^2_{\mathrm{res}}, \qquad \mathbf{g}_* = \left(1, \frac{t_*}{\hat{\beta}_2 + t_*}, -\frac{\hat{\beta}_1\, t_*}{(\hat{\beta}_2 + t_*)^2}\right)^\top.$$

Here $\hat{\sigma}^2_{\mathrm{res}}$ denotes the residual variance of the fitted curve, with divisor $n-3$ for n observations. The predictive distribution is approximated by a Gaussian distribution centred at the fitted trajectory and using $\sigma^2_{\mathrm{pred}}(t_*)$ as its variance.

The member reverts to the pre-specified local-linear fallback if fewer than eight observations are available, if the observed range of the history is at most 0.01 mm, or if the bounded

nonlinear fit fails or returns a non-finite parameter covariance. The fallback fits a least-squares line to the 15 most recent observations, or to all of them when fewer are available, extrapolates it, and uses the predictive standard deviation $\hat{s}\sqrt{1+h}$ at lead h, where $\hat{s}$ is the larger of the residual standard deviation of the line and the robust scale of the increments of these observations. The same fallback convention is used whenever another Gaussian member cannot issue a valid forecast. The fitting bounds and minimum-history rules are given in Table S1, in which the fitted curve is written $y = a + bt/(c + t)$ and (a, b, c) corresponds to $(\beta_0, \beta_1, \beta_2)$.

LightGBM quantile-regression model. The global LightGBM member is trained on pooled source-project forecast–target pairs with increment response $z_i = x_{t_i+h_i} - x_{t_i}$. For each α in the common grid $\mathcal{A}$ defined in the main text, a separate gradient-boosted model is fitted by minimising the pinball loss $\sum_i \rho_\alpha(z_i - \hat{z}_\alpha(\mathbf{x}_i))$, where $\rho_\alpha(u) = u\{\alpha - \mathbf{1}(u < 0)\}$ and $\mathbf{x}_i$ is the feature vector of training pair i.

The feature vector $\mathbf{x}_{j,t,h}$ contains the current recorded level x_t, the one-, three-, and five-session increments, the average rate over the five most recent session intervals, the robust scale of the seven most recent increments, elapsed monitoring time since the first session, the three-session groundwater change, forecast lead, and instrument family. At a forecast origin all temporal features are constructed from the history up to that origin. In a training row the displacement and time features are constructed relative to the origin of that row, whereas the groundwater change is the change over the last three sessions of the same series in the data used for the fit, set to zero when either reading is missing. As a result, no observation after the refitting date enters a training row, and the groundwater change is constant within each series of a refit but can differ between series. In several refits the fitted models depend on this feature. The measurement-scale quantile is $Q_{\mathrm{LightGBM},j,t,h}(\alpha) = x_t + \hat{z}_\alpha(\mathbf{x}_{j,t,h})$. The fifteen independently fitted quantiles are monotonically rearranged when necessary.

On Project A, the global LightGBM models are refitted every ten calendar days using only observations available before the refitting date. Their number of leaves, learning rate, boosting rounds, minimum leaf size, sampling fractions, and other fixed hyperparameters are listed in Table S1. The transferred Project-B models are frozen at their source-project values. Because Project B has no groundwater record, its groundwater-change feature is zero at every origin.

GRU ensemble. The second global member represents recent sequential behaviour. At each origin, the most recent $L_{\mathrm{seq}} = 16$ increments are encoded as $\mathbf{u}_k = ((x_k - x_{k-1})/(\Delta t_k\, s_{j,t}), \log(1 + \Delta t_k), v_k)$, where Δt_k is the elapsed time between successive observations and v_k is a validity indicator. For histories shorter than L_{seq}, the available increment steps are right-aligned within the fixed-length input, preceding positions are zero-padded, and $v_k = 1$ for observed steps and $v_k = 0$ for padded positions. The same sequence construction is used in source-project training and Project-B deployment, except for the scale. At a forecast origin the input increments and the predicted increment are scaled by $s_{j,t}$, whereas the inputs and the target of every training row of a fit are scaled by the robust scale of all increments of its series in the data used for that fit. The recurrent state is combined with the normalised lead $h/7$ and the instrument-family indicator to predict the scaled future increment $z = (x_{t+h} - x_t)/s_{j,t}$.

Five independently initialised replicas, indexed by $r = 1, \ldots, R$ with $R = 5$, are trained by mean squared error (MSE). If $\hat{z}^{(r)}$ denotes the prediction of replica r, the predictive location

is the ensemble mean and the predictive variance combines between-replica dispersion with the training residual variance σ^2_{res} of the scaled increment for this member:

$$\mu_z = R^{-1} \sum_{r=1}^{R} \hat{z}^{(r)}, \qquad \sigma_z^2 = \mathrm{Var}_r\{\hat{z}^{(r)}\} + \sigma^2_{\mathrm{res}},$$

where Var_r denotes the empirical variance, with divisor R, over the R replica predictions, and σ^2_{res} is the variance, with divisor $n - 1$, of the residuals of the ensemble mean over the n training rows of the fit, pooled across leads. The moments μ_z and σ_z are transformed back as $\mu = x_t + s_{j,t}\mu_z$ and $\sigma = s_{j,t}\sigma_z$. The network has one GRU layer and is trained for a fixed number of epochs without early stopping. The hidden size, the dimensions of the fully connected output head and its rectified linear unit (ReLU) activation, the optimiser, the learning rate, the batch size, and the number of epochs are summarised in Table S1.

Common probabilistic output. For the LLT, GP, rectangular-hyperbola, and GRU members, the Gaussian predictive moments are converted to quantiles as $Q_{m,j,t,h}(\alpha) = \mu_{m,j,t,h} + \sigma_{m,j,t,h}\Phi^{-1}(\alpha)$, where Φ denotes the standard normal CDF and $\mu_{m,j,t,h}$ and $\sigma_{m,j,t,h}$ are the predictive mean and standard deviation obtained above for member m, with the predictive standard deviation bounded below by 0.05 mm. LightGBM supplies its quantiles directly. All members are evaluated on the common grid $\mathcal{A}$ defined in the main text, followed by monotone rearrangement and linear interpolation. No member-specific tail completion is applied at this stage. The members enter the pool through their interior distributions, and the common tail-completion rule is applied only when a pooled or member distribution is scored or converted to probabilities.

S3. Probability-integral-transform flag census

A forecast–observation pair receives a PIT flag and is marked for review when its PIT under the completed distribution falls outside $[0.002, 0.998]$ (soft flag), and a flag whose PIT lies within 10^{-6} of 0 or 1 is a hard flag. These thresholds define a screening diagnostic.

Among the scored pairs, Project A contains 33 flagged pairs, 25 on the held-out window under the issued distributions and the others on the training window under the static prior, including 1 hard flag at crown-beam monitoring point JC7 for the three-day forecast issued on 6 May, whose observation lies in the upper tail of the issued distribution. Project B contains 6 flagged pairs and no hard flag.

A separate data-quality case is the pair of corrected level-shift artifacts recorded on 3 April 2026 at road monitoring points D7-2 and D7-3, whose raw readings imply one-step jumps of −79 mm and −87 mm. At D7-3 the raw reading of −89 mm receives a numerically zero PIT, whereas the audit-corrected reading has PIT 0.24 and remains consistent with the predictive band. The monitoring record shows contemporaneous per-session changes of only 0.03 mm at D7-2 and −0.05 mm at D7-3, which supports treating the two jumps as measurement artifacts rather than deformation.

S4. Per-series modelling roles

Table S2 records the modelling role of each Project-A series. The role codes are T for a source-project forecast target used in member training, C for entry into pre-deployment ACP-

Table S1: Implementation and hyperparameter settings of the five probabilistic forecasting members.

Member	Specification
Local linear trend	`statsmodels` UnobservedComponents, state-space form on the daily grid (missing days filtered). The three variances are estimated once per series by ML (maxiter 200) on its history at the first forecast origin, that is, the burn-in window and the reading at that origin, and kept fixed, with Gaussian forecasts.
Gaussian process	`scikit-learn` GPR on the last 60 observations, kernel $\mathrm{Const}(1.0, [0.05, 50])\times\mathrm{RBF}(10\,\mathrm{d}, [2, 90]) + \mathrm{White}(0.3, [10^{-3}, 20])$, targets standardised by the robust scale, one optimiser restart.
Rectangular hyperbola	$y = a + bt/(c + t)$ by nonlinear least squares (bounds $\lvert a\rvert, \lvert b\rvert \le 500$, $c \in [1, 5{,}000]$), delta-method predictive variance plus residual variance. Local-linear fallback when under 8 points, when the series range is at most 0.01 mm, or when the fit fails or returns a non-finite parameter covariance.
LightGBM	pooled quantile regression, one model per level: 250 trees, learning rate 0.06, 31 leaves, min. 25 samples/leaf, subsample 0.9, feature fraction 0.9. Features: current recorded level x_t, increments d_1, d_3, d_5, average rate over the last five session intervals, robust scale of the last seven increments, elapsed days, 3-session water change, h, instrument-family flag. Monotone rearrangement of predicted quantile vectors, 10-day refits.
GRU ensemble	five GRU(32) networks with a 34-32-1 ReLU head (input: 16 steps of scaled per-day increment, log gap, validity flag), seeds 100–104, Adam, learning rate $2 \cdot 10^{-3}$, 30 epochs, batch 512, MSE on the scaled increment, predictive variance = ensemble spread + training residual variance, Gaussian quantiles.
All members	15 output levels $\mathcal{A}$, scale floor 0.05 mm. Gaussian process and hyperbola refitted per origin, local-linear-trend state re-filtered with fixed variances, global members refit every 10 days on data before the origin.

prior selection, E for entry into the held-out Project-A forecast evaluation, S for a system-layer marginal, A for the anomaly audit, and X for an auxiliary global-member forecast used only in system-layer assessment. The road and building settlement series used at the system layer do not enter ACP-prior selection or the Project-A forecast score.

S5. ACP-prior selection

The QA and SLP families are optimised separately with the non-dominated sorting genetic algorithm II (NSGA-II) on the 5,234 Project-A training pairs using the three objectives defined in the main text. Their Pareto fronts from the first optimiser seed are then combined, and the coverage-prioritised rule in Eq. (5) of the main text is applied to the resulting candidate set. Because no point of the candidate set attains the group-coverage floor of 0.85 on the training window, the selected candidate is the point with the smallest worst-group coverage shortfall, with mean training CRPS used only to resolve candidates attaining the same minimum shortfall.

The resulting ACP prior has $\kappa^* = \mathrm{SLP}$ and $\tau^* = 1.69$. Its member weights $\mathbf{w}^*$, together with the principal reference operating points, appear in Table S3. The knee point of each family is the Pareto-front point closest to the ideal point after min–max normalisation of the three objectives. The minimum-CRPS point is the QA front point with the lowest mean training CRPS. The scalar-CRPS optimum is obtained by minimising mean training CRPS

Table S2: Per-series modelling roles on Project A.

Series	Family	Roles
S3	inclinometer	T, C, E, S, A
S4	inclinometer	T, C, E, S, A
S5	inclinometer	T, C, E, S, A
S6	inclinometer	T, C, E, S, A
SC1	inclinometer	T, C, E, S, A
SC2	inclinometer	T, C, E, S, A
JC1	crown	T, C, E, S, A
JC2	crown	T, C, E, S, A
JC3	crown	T, C, E, S, A
JC4	crown	T, C, E, S, A
JC6	crown	T, C, E, S, A
JC7	crown	T, C, E, S, A
JC8	crown	T, C, E, S, A
JC9	crown	T, C, E, S, A
JC10	crown	T, C, E, S, A
JC11	crown	T, C, E, S, A
JC12	crown	T, C, E, S, A
JC13	crown	T, C, E, S, A
JC14	crown	T, C, E, S, A
JC15	crown	T, C, E, S, A
JC16	crown	T, C, E, S, A
JC17	crown	T, C, E, S, A
JC18	crown	T, C, E, S, A
JC19	crown	T, C, E, S, A
JC20	crown	T, C, E, S, A
JC21	crown	T, C, E, S, A
LB6	building	S, A, X
LB11	building	S, A, X
LB18	building	S, A, X
LB31	building	S, A, X
D5-1	road	S, A, X
D6-1	road	S, A, X
D7-1	road	S, A, X
D7-2	road	S, A, X
D7-3	road	S, A, X
D11-2	road	S, A, X

directly, as a single objective over the weights and the common spread factor of the QA family, with the Nelder–Mead method started from equal weights and $\tau = 1$.

Table S3: Pooling weights and spread factors of the ACP prior and principal reference operating points.

Operating point	Family	LLT	GP	Hyperbola	LightGBM	GRU	τ
ACP prior (coverage-prioritised union point)*	SLP	0.31	0.05	0.00	0.58	0.05	1.69
SLP knee point	SLP	0.40	0.03	0.01	0.40	0.15	0.82
QA knee point	QA	0.32	0.14	0.02	0.50	0.03	0.98
Min-CRPS point	QA	0.23	0.10	0.01	0.51	0.15	1.50
Scalar-CRPS optimum	QA	0.26	0.12	0.06	0.41	0.16	1.48

Note: * marks the operating point selected for deployment. Member labels are defined in Section S2.

S6. Stratified calibration diagnostics

The held-out Project-A forecasts are stratified by forecast lead and by instrument family, the two dimensions whose crossing defines the eligible groups $\mathcal{G}_{30}$ in the calibration objective of the main text. Table S4 gives the test scores of the ACP prior, the LightGBM member, and the equal-weight pool, so that calibration can be checked lead by lead and family by family.

Because the raw group-wise PIT CvM discrepancy depends on group size, we also compute the descriptive standardisation $W_g^2 = N_g \, \mathrm{CvM}_g + 1/(12N_g)$, where CvM_g denotes the group-g PIT CvM discrepancy entering the calibration objective of the main text and N_g is the number of eligible pairs in group g. This standardisation is the classical CvM statistic of the group PIT values and puts the groups on a common scale. We do not apply the classical critical values, because the rolling forecasts are serially dependent. The worst held-out value for the

Table S4: Stratified test scores by forecast lead and instrument family.

Stratum	n pairs	Method	CRPS	CvM	Cov. 80 %	Cov. 90 %
$h = 1$	780	ACP prior	0.605	0.0026	0.94	0.99
		LightGBM	0.611	0.0032	0.64	0.75
		Equal-weight	0.721	0.0008	0.82	0.93
$h = 3$	754	ACP prior	0.972	0.0024	0.90	0.97
		LightGBM	0.966	0.0021	0.67	0.81
		Equal-weight	1.040	0.0008	0.75	0.85
$h = 5$	988	ACP prior	1.107	0.0032	0.91	0.97
		LightGBM	1.101	0.0027	0.64	0.77
		Equal-weight	1.195	0.0006	0.75	0.85
$h = 7$	728	ACP prior	1.450	0.0019	0.88	0.95
		LightGBM	1.441	0.0031	0.63	0.74
		Equal-weight	1.538	0.0017	0.68	0.80
crown	2,500	ACP prior	1.070	0.0026	0.90	0.97
		LightGBM	1.085	0.0029	0.64	0.76
		Equal-weight	1.123	0.0059	0.74	0.84
inclinometer	750	ACP prior	0.905	0.0078	0.93	0.98
		LightGBM	0.841	0.0064	0.67	0.81
		Equal-weight	1.117	0.0434	0.81	0.92

Note: All strata use the same held-out forecast origins.

ACP prior is 2.431 in group $h = 5$, crown. The two family knee points and the equal-weight, local-only, and global-only pools have higher worst held-out values, from 2.710 to 15.608.

Figure S1 shows the PIT histograms of the five members and the ACP prior with dependence-aware moving-block-bootstrap bands. The histograms pool the training and held-out windows.

S7. Dependence diagnostics

Cross-instrument dependence is estimated from PIT values of resolved forecast–observation pairs rather than from the raw monitoring levels. Each series is oriented towards its alarm limit, as in the exceedance events that the copula layer couples. Pairwise-complete Spearman correlations ρ_S are mapped to Gaussian-copula correlations $2\sin(\pi\rho_S/6)$. The resulting correlation matrix is shrunk towards an exchangeable target of 0.3 with a variance-weighted intensity and repaired, when necessary, to a valid correlation matrix by flooring its eigenvalues at 10^{-4} and renormalising its diagonal to one. On the Project-A panel at the three-day lead, the estimated weight retained on the empirical matrix is 0.89, the remainder being placed on the target. On Project B, whose instrument pairs share a median of 13 overlapping resolved pairs at the three-day lead (Table S5), the point estimate of the system-level exceedance probability instead uses an exchangeable correlation matrix whose common coefficient is the mean pairwise estimate at that lead before shrinkage, 0.10, computed once from all resolved replay pairs. The same coefficient also couples the instruments whose larger probability is that of the next-day rate event. The one-day and three-day estimates in Table S5 lie inside each other's 95% confidence intervals on both projects, consistent with a coefficient common to the two leads and channels. Shrinking the Project-B matrix towards zero or towards 0.3 gives mean correlations of 0.07 and 0.15, and the corresponding exchangeable

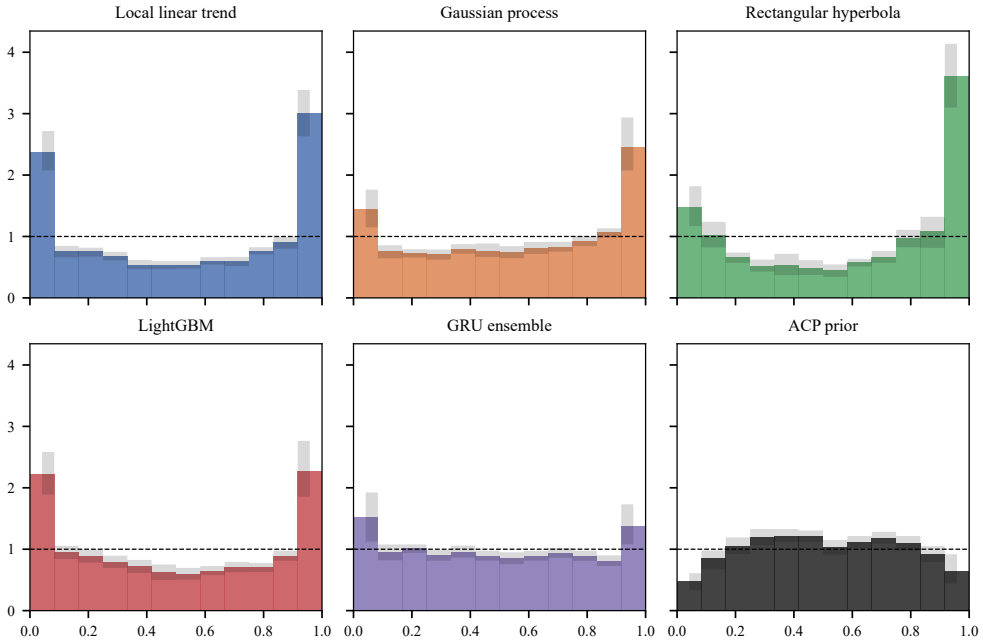


Figure S1: PIT histograms of the five members and the ACP prior for all Project-A rolling forecasts, with moving-block-bootstrap bands.

Gaussian probabilities lie inside the span of the parametric scenarios. Table S5 presents the mean pairwise dependence estimates before shrinkage, the minimum and median numbers of overlapping resolved pairs per instrument pair, and moving-block-bootstrap 95% confidence intervals (CI) obtained by resampling forecast-origin dates in blocks of five consecutive dates on both projects, with 300 replicates.

Table S5: Cross-instrument dependence diagnostics for the copula layer.

Project	Horizon	Empirical $\bar{\rho}$	95 % CI	Overlap (min/med)
A	$h = 1$	0.02	[0.00, 0.04]	69/82
A	$h = 3$	0.03	[0.00, 0.05]	65/79
B	$h = 1$	0.06	[−0.02, 0.17]	15/15
B	$h = 3$	0.10	[−0.04, 0.37]	13/13

Note: Empirical $\bar{\rho}$ is the mean pairwise dependence estimate before shrinkage.

Among the parametric scenarios, the exchangeable Gaussian copula at 0.10 uses the estimated association. The other parametric scenarios are exchangeable Gaussian copulas with correlations 0.15, 0.30, 0.45, and 0.60, exchangeable Student-t copulas with correlation 0.30 and 3, 4, or 8 degrees of freedom, and independence. The Gaussian and Student-t copula probabilities are Monte Carlo estimates from 20,000 joint draws, with a standard error of 0.001 at the peak of the estimated Gaussian scenario. The Fréchet bounds in the main text are the primary dependence-robust system statement.

S8. Event-level warning audit

Event eligibility is fixed at the first evaluable origin of each Project-B series. The horizontal series are first forecast at the origin of 24 April and the settlement series at the origin of 26 April, because the settlement baseline was established later. W10H had crossed its cumulative limit on 23 April, before the first horizontal origin, and W13V had crossed its settlement limit on 26 April, at the first settlement origin. Both are therefore ineligible for prospective cumulative-warning evaluation and excluded from cumulative-event detection and from the calculation of warning lead times. W12V crossed its 50 mm settlement limit on 28 April and is the prospectively evaluable cumulative event. Table S6 records the per-series observation window, first evaluable forecast origin, crossing date, and the instrument-day denominators of both channels. Obs. and First obs. give the number of observations and the first observation date. Cum. days gives the three-day cumulative-channel instrument-days, and At-risk days gives the subset of them on which the origin reading has not yet crossed the cumulative limit. Rate days gives the one-day rate-channel instrument-days, and Rate pos. gives the number of them with an observed daily increment at or above the rate limit.

For the cumulative channel, a warning is issued when $p_j^{\mathrm{cum}}(t, 3) \geq p^*$. A warning qualifies when it is issued within the three days before the first observed crossing, and warning lead time is measured from the issue date of the earliest qualifying warning to the first observed crossing date. At the primary trigger $p^* = 0.35$, W12V is first warned on 27 April, when its oriented reading is 47.4 mm, and crosses on 28 April, giving the one-day warning lead time of the main text.

For instrument-day bookkeeping, the cumulative endpoint label is $\mathbf{1}\{X_j(t+3) \geq L_j\}$ and the one-day rate label is $\mathbf{1}\{X_j(t+1) - x_j(t) \geq r_j\}$. A series leaves the at-risk set of the cumulative channel once its origin reading has crossed L_j, whereas the rate channel retains every one-day pair, including those that follow a cumulative crossing. Warned event-negative days on the same instrument form one false-warning episode when successive such days are at most two calendar days apart, so a single intervening day without a false warning does not end an episode. Alarm burden is the fraction of evaluable instrument-days on which a warning is issued. At $p^* = 0.35$, the deployed ACP records one detected cumulative event, 2 false-warning episodes, and an alarm burden of 0.03. Table S7 contains the underlying instrument-day counts of both channels. A stable series never crosses its limit and remains below $0.95L_j$ throughout the replay, and a near-limit series peaks in $[0.95L_j, L_j)$ without crossing. The true-positive (TP) and false-negative (FN) counts are the warned and the missed event-positive instrument-days, and the false-positive rate (FPR) is the warned fraction of event-negative instrument-days, computed over stable instrument-days for FPR stable and over all event-negative instrument-days for FPR all. Near-limit warn divides warned near-limit instrument-days by their total, and TP all days counts true-positive days including those after the first observed crossing.

The rate channel is evaluated separately. A daily-rate episode of a series is a maximal run of consecutive days, after its first forecast origin, on which the observed daily increment is at or above r_j, so increments observed up to that origin do not form episodes. Daily-rate episodes serve as a secondary short-horizon warning diagnostic, and the warning lead time is computed from the cumulative crossing alone. Table S8 lists every rate episode with its first-day increment, the increment of the preceding day, the first increment divided by the

Table S6: Per-series observation window, eligibility, and instrument-day denominators of the Project-B replay.

Series	Obs.	First obs.	First origin	Crossing	Eligible	Cum. days	At-risk days	Rate days	Rate pos.
W9H	23	04-18	04-24	–	yes	15	15	17	0
W10H	23	04-18	04-24	04-23	no	15	0	17	17
W11H	23	04-18	04-24	–	yes	15	15	17	7
W12H	23	04-18	04-24	–	yes	15	15	17	6
W13H	23	04-18	04-24	–	yes	15	15	17	2
W9V	21	04-21	04-26	–	yes	13	13	15	0
W10V	21	04-21	04-26	–	yes	13	13	15	2
W11V	21	04-21	04-26	–	yes	13	13	15	6
W12V	21	04-21	04-26	04-28	yes	13	2	15	11
W13V	21	04-21	04-26	04-26	no	13	0	15	5

Note: Dates are month–day in 2021. Eligible marks series whose crossing, if any, follows the first evaluable origin.

Table S7: Instrument-day counts underlying the Project-B warning metrics.

Channel	p^*	TP	FN	FPR stable	FPR all	Near-limit warn	TP all days
Cumulative	0.20	2	0	0.07	0.17	12/30	41
Cumulative	0.35	1	1	0.01	0.02	1/30	40
Cumulative	0.50	0	2	0.00	0.00	0/30	39
Rate	0.20	44	12	0.35	0.50	25/34	44
Rate	0.35	32	24	0.20	0.23	7/34	32
Rate	0.50	14	42	0.14	0.13	1/34	14

Note: Cumulative rows use the $h = 3$ endpoint label on the at-risk set. Rate rows use the $h = 1$ label.

robust scale at the origin, the issued one-day rate-exceedance probability on the first day and its maximum within the episode, and its detection at the primary trigger and at the lower trigger of 0.20. An episode counts as detected at a trigger when the rate warning is issued for at least one of its days. The preceding-day increment is shown as a dash for episodes that start on the day after the first forecast origin of the series, because no one-day replay pair ends on that origin.

S9. Ablation and sensitivity analyses

All analyses reuse the Project-B replay protocol and the stored member forecasts, except those of the alternative global member, which is trained with the protocol of the GRU ensemble. The online layer keeps the deployed adaptation constants unless a table or the text states otherwise. Where a member set or a scale definition changes the pre-deployment selection, both family fronts are re-optimised and the coverage-prioritised rule is re-applied so that every alternative configuration has been selected on Project-A training data only.

Member sets and selection rule. Table S9 places the deployed ACP and the static ACP prior alongside matched removals of one member from the deployed prior with renormalised weights under the deployed online layer. It also includes, with and without that layer, the scalar-CRPS and minimum-CRPS operating points and the member subsets carried through the full selection procedure described above, labelled full proc. in the table. Its column labels extend those of Table 6 of the main text, and the later ablation tables reuse them. A CRPS and A C.90 are the mean scale-normalised CRPS and the empirical coverage of the central

Table S8: Episode-level audit of the daily-rate channel on Project B for the deployed ACP.

Series	Start	Days	First inc. (mm)	Prev. day (mm)	Inc./scale	p^{rate} first	p^{rate} max	Warned 0.35/0.20
W10H	04-25	17	5.20	–	4.38	0.31	0.48	1/1
W11H	04-26	2	2.80	1.50	1.45	0.18	0.18	0/0
W11H	05-02	1	2.80	−0.90	1.18	0.60	0.60	1/1
W11H	05-05	2	3.00	0.80	1.56	0.49	0.49	1/1
W11H	05-09	1	4.00	1.60	2.25	0.34	0.34	0/1
W11H	05-11	1	3.60	1.40	2.02	0.18	0.18	0/0
W12H	04-25	2	3.50	–	1.31	0.19	0.19	0/0
W12H	04-28	1	3.10	−1.00	0.89	0.25	0.25	0/1
W12H	04-30	2	3.40	−1.60	1.04	0.56	0.56	1/1
W12H	05-11	1	4.00	0.50	1.80	0.68	0.68	1/1
W13H	04-25	2	5.50	–	2.06	0.30	0.30	0/1
W10V	05-05	1	2.90	0.40	4.89	0.08	0.08	0/0
W10V	05-10	1	3.00	0.10	6.74	0.08	0.08	0/0
W11V	04-27	2	4.50	–	15.18	0.17	0.34	0/1
W11V	05-05	1	2.90	1.60	3.91	0.14	0.14	0/0
W11V	05-09	3	2.10	−0.40	2.83	0.06	0.58	1/1
W12V	04-27	2	11.80	–	79.59	0.36	0.37	1/1
W12V	04-30	1	3.40	1.50	1.70	0.64	0.64	1/1
W12V	05-02	6	3.50	1.40	1.15	0.97	0.99	1/1
W12V	05-10	2	6.00	1.60	2.45	0.96	0.96	1/1
W13V	04-27	2	5.20	–	7.01	0.38	0.46	1/1
W13V	05-05	1	2.10	1.30	0.59	0.20	0.20	0/1
W13V	05-10	2	2.80	0.10	1.51	0.17	0.21	0/1

Note: Inc./scale is the first increment divided by the robust scale. Warned marks detection at the primary and lower trigger.

90% interval on the Project-A held-out window, and B CRPS and B C.90 are the same scores on Project B. A w.stg is the worst-stage scale-normalised CRPS on the Project-A held-out window, and B C.90 H and B C.90 V split the Project-B coverage into the horizontal and settlement series. The event columns are evaluated at $p^* = 0.35$. Lead is the warning lead time in days, with a dash where no warning precedes the crossing. F.ep. is the number of false-warning episodes, Burden is the alarm burden of the cumulative channel, and Rate det. is the number of daily-rate episodes detected. Table S10 lists the operating points selected for each member subset.

Table S9: Member-set and selection-rule ablations with and without the deployed online layer.

Configuration	A CRPS	A C.90	A w.stg	B CRPS	B C.90	B C.90 H	B C.90 V	Lead	F.ep.	Burden	Rate det.
ACP (deployed)	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
ACP prior, static	1.032	0.97	1.056	1.879	0.93	0.93	0.94	+1	1	0.02	5
ACP prior without hyperbola + online	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
ACP prior without GRU + online	1.014	0.88	1.041	1.846	0.85	0.85	0.84	+1	2	0.03	12
ACP prior without LightGBM + online	1.070	0.91	1.102	1.700	0.92	0.91	0.92	+2	2	0.22	17
Scalar-CRPS opt., static	1.031	0.96	1.054	1.678	0.85	0.86	0.83	+1	3	0.09	10
Scalar-CRPS opt. + online	1.005	0.88	1.033	1.720	0.84	0.87	0.82	+1	2	0.12	13
Min-CRPS point, static	1.013	0.95	1.037	1.828	0.78	0.79	0.76	+1	3	0.05	8
Min-CRPS point + online	1.001	0.88	1.029	1.850	0.82	0.83	0.80	+1	2	0.12	12
Local-only, full proc., static	1.146	0.98	1.166	1.540	0.97	0.97	0.97	+2	2	0.18	20
Local-only, full proc. + online	1.122	0.90	1.155	1.519	0.91	0.93	0.89	+2	2	0.16	18
GRU removed, full proc., static	1.044	0.97	1.068	1.666	0.75	0.77	0.73	+1	3	0.07	9
GRU removed, full proc. + online	1.017	0.87	1.044	1.863	0.82	0.85	0.79	+1	1	0.12	13
LightGBM removed, full proc., static	1.081	0.97	1.103	1.627	0.97	0.97	0.97	+2	2	0.15	18
LightGBM removed, full proc. + online	1.036	0.91	1.066	1.631	0.91	0.90	0.91	+2	3	0.15	17
Global-only, full proc., static	1.048	0.98	1.072	2.839	0.42	0.48	0.34	–	0	0.00	0
Global-only, full proc. + online	1.010	0.89	1.037	2.688	0.80	0.85	0.76	–	0	0.00	0

Note: Columns follow Table 6 of the main text. H and V denote the horizontal and settlement series.

Table S10: Operating points selected by the full procedure on each member subset.

Member set	Family	τ	LLT	GP	Hyperbola	LightGBM	GRU	Min. cov.
Local-only	SLP	1.53	0.56	0.31	0.13	0.00	0.00	0.874
GRU removed	QA	1.79	0.33	0.02	0.02	0.62	0.00	0.855
LightGBM removed	SLP	1.58	0.53	0.06	0.03	0.00	0.38	0.828
Global-only	SLP	1.80	0.00	0.00	0.00	0.86	0.14	0.845

Note: Min. cov. is the worst-group training-window coverage of the central 90% interval.

Online updates and buffer length. Table S11 separates the two online updates and varies the buffer length W. In the row with the flagged pairs excluded, the 33 Project-A and 6 Project-B pairs flagged by the PIT screening of Section S3 are kept out of the weight and spread updates but are still issued and scored. The paired CRPS intervals against the deployed ACP, $[-0.002, 0.0002]$ on Project A and $[-0.059, 0.018]$ on Project B, include zero, and the warning outcomes are unchanged.

Table S11: Online-layer ablation and buffer-length sensitivity.

Configuration	A CRPS	A C.90	A w.stg	B CRPS	B C.90	B C.90 H	B C.90 V	Lead	F.ep.	Burden	Rate det.
No online update	1.032	0.97	1.056	1.879	0.93	0.93	0.94	+1	1	0.02	5
Weight update only	1.027	0.97	1.051	1.814	0.94	0.93	0.96	+1	4	0.08	12
Spread update only	1.014	0.88	1.043	1.934	0.84	0.82	0.86	+1	1	0.02	6
Both updates (deployed)	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
Flagged pairs excluded from updates	1.007	0.88	1.035	1.829	0.85	0.87	0.83	+1	2	0.03	11
Buffer $W = 15$	1.009	0.87	1.037	1.837	0.84	0.86	0.83	+1	2	0.03	11
Buffer $W = 25$	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
Buffer $W = 50$	1.004	0.90	1.032	1.851	0.86	0.86	0.86	+1	3	0.04	11
Buffer $W = 100$	1.002	0.90	1.029	1.843	0.90	0.89	0.91	+1	3	0.04	11

Anchoring strength. Table S12 varies the anchoring strength γ, including the unanchored case, and shows in its last column the smallest combined deployed weight of the two global members during the Project-B replay. Table S13 lists the validation CRPS of the pre-specification grid of the adaptation constants together with its extension to $\gamma \in \{0, 0.6, 0.8\}$ and $W = 50$. The fully non-anticipative variant of Section 5.5 of the main text selects $(\eta, \gamma, W) = (0.2, 0.1, 15)$ from the pre-specification grid and re-selects the prior. Each of these two changes to the deployed ACP is applied alone to separate its effect. With the deployed prior, these constants give a CRPS and coverage of 1.010 and 0.87 on Project A and 1.898 and 0.84 on Project B (paired interval $[-0.004, 0.116]$), with 5 false-warning episodes and an alarm burden of 0.14. The prior re-selected on the 4,818 resolved training pairs, with the deployed constants, gives 1.016 and 0.88 on Project A and 1.816 and 0.85 on Project B (paired interval $[-0.211, 0.073]$), with 3 false-warning episodes and an alarm burden of 0.06.

Coverage floor. The deployment rule is re-applied to the stored union front for alternative coverage floors (Table S14).

Member-specific spread factors. Table S15 compares the common spread factor with member-specific factors τ_m selected by the same procedure.

Table S12: Sensitivity to the anchoring strength γ.

γ	A CRPS	A C.90	B CRPS	B C.90	B C.90 H	B C.90 V	Lead	F.ep.	Min. global weight on B
0.0	1.009	0.88	1.940	0.87	0.87	0.87	+1	4	0.05
0.2	1.008	0.88	1.876	0.85	0.86	0.83	+1	4	0.17
0.4	1.008	0.88	1.848	0.85	0.86	0.83	+1	2	0.28
0.6	1.009	0.88	1.859	0.83	0.85	0.80	+1	1	0.40
0.8	1.011	0.88	1.873	0.84	0.81	0.86	+1	1	0.52
1.0	1.014	0.88	1.934	0.84	0.82	0.86	+1	1	0.63

Note: $\gamma = 1$ fixes the deployed weights at the ACP prior while the spread update remains active.

Table S13: Validation CRPS of the adaptation-constant grid on the inner validation window.

$\eta \setminus \gamma$	0.0	0.1	0.2	0.4	0.6	0.8
0.2	1.130 (50)	1.128 (50)	1.127 (50)	1.125 (50)	1.123 (50)	1.122 (50)
0.5	1.148 (50)	1.144 (50)	1.140 (50)	1.131 (50)	1.126 (50)	1.123 (50)
1.0	1.194 (50)	1.180 (50)	1.168 (50)	1.149 (50)	1.134 (50)	1.125 (50)

Note: Cells give the best CRPS over $W \in \{15, 25, 50\}$ (W in parentheses). The deployed $(\eta, \gamma, W) = (0.2, 0.4, 25)$, best on the pre-specification grid, scores 1.128.

Table S14: Sensitivity of the deployed operating point to the coverage floor.

Floor	Family	τ	LLT	GP	Hyperbola	LightGBM	GRU	Feasible	Min. cov.	A CRPS	A C.90	B CRPS	B C.90	Lead	F.ep.
0.70	SLP	1.45	0.26	0.04	0.02	0.48	0.19	47	0.800	1.003	0.89	1.801	0.86	+1	2
0.75	SLP	1.45	0.26	0.04	0.02	0.48	0.19	33	0.800	1.003	0.89	1.801	0.86	+1	2
0.80	SLP	1.45	0.26	0.04	0.02	0.48	0.19	15	0.800	1.003	0.89	1.801	0.86	+1	2
0.83	SLP	1.45	0.30	0.08	0.06	0.44	0.11	8	0.831	1.007	0.89	1.690	0.87	+1	3
0.85	SLP	1.69	0.31	0.05	0.00	0.58	0.05	0	0.848	1.008	0.88	1.848	0.85	+1	2
0.88	SLP	1.69	0.31	0.05	0.00	0.58	0.05	0	0.848	1.008	0.88	1.848	0.85	+1	2
0.90	SLP	1.69	0.31	0.05	0.00	0.58	0.05	0	0.848	1.008	0.88	1.848	0.85	+1	2
0.95	SLP	1.69	0.31	0.05	0.00	0.58	0.05	0	0.848	1.008	0.88	1.848	0.85	+1	2

Note: Feasible counts union-front points whose worst-group training-window coverage reaches the floor.

Tail completion. We recompute the Project-B warning metrics and the Project-A probability validation at the series-specific 80th- and 90th-percentile evaluation thresholds (P_{80} and P_{90}) under alternative tail families (Table S16). The last column of the table gives the largest probability of exceeding a governing threshold among the 754 held-out three-day Project-A forecasts. This far-tail probability ranges from 0 without tail mass beyond the stored quantiles to 2.7×10^{-4} under the Student-t_3 tail, whereas the warning metrics, recomputed from the same issued distributions, are identical under every tail family.

Robust scale. Table S17 varies the definition and floor of the robust scale used by the scoring and online layers, including a series-constant convention computed over the complete monitoring record.

Table S15: Common versus member-specific spread factors.

Configuration	Family	τ_{LLT}	τ_{GP}	$\tau_{\text{Hyperbola}}$	τ_{LightGBM}	τ_{GRU}	Min. cov.	A CRPS / C.90	B CRPS / C.90
Common τ (deployed)	SLP	1.69	1.69	1.69	1.69	1.69	0.848	1.008 / 0.88	1.848 / 0.85
Member-specific τ_m	SLP	1.75	1.59	1.55	1.72	0.62	0.852	1.011 / 0.88	1.864 / 0.85

Table S16: Tail-family sensitivity of the warning metrics and of the P_{80}/P_{90} evaluation-threshold validation.

Tail family	p^* = 0.10			p^* = 0.20			p^* = 0.35			p^* = 0.50			Brier		A limit
	Lead	F.ep.	Rate det.	Lead	F.ep.	Rate det.	Lead	F.ep.	Rate det.	Lead	F.ep.	Rate det.	P_{80}	P_{90}	max. prob.
Gaussian (deployed)	+2	2	21/23	+2	2	17/23	+1	2	11/23	–	0	7/23	0.088	0.086	$< 10^{-15}$
Exponential	+2	2	21/23	+2	2	17/23	+1	2	11/23	–	0	7/23	0.088	0.086	8.8×10^{-7}
Student-t_3	+2	2	21/23	+2	2	17/23	+1	2	11/23	–	0	7/23	0.088	0.086	2.7×10^{-4}
Student-t_5	+2	2	21/23	+2	2	17/23	+1	2	11/23	–	0	7/23	0.088	0.086	3.8×10^{-5}
No tail mass beyond $q_{0.99}$	+2	2	21/23	+2	2	17/23	+1	2	11/23	–	0	7/23	0.088	0.086	0

Table S17: Sensitivity to the robust-scale definition.

Configuration	A CRPS	A C.90	A w.stg	B CRPS	B C.90	B C.90 H	B C.90 V	Lead	F.ep.	Burden	Rate det.
MAD, floor 0.05 (deployed)	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
MAD, floor 0.02	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
MAD, floor 0.10	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
IQR/1.349, floor 0.05	1.018	0.88	1.045	1.549	0.84	0.86	0.82	+1	1	0.02	10
IQR/1.349, re-selected prior	1.018	0.88	1.045	1.526	0.86	0.89	0.83	+1	1	0.02	10
SD, floor 0.05	1.051	0.88	1.079	1.042	0.85	0.83	0.87	+1	1	0.02	9
SD, re-selected prior	1.057	0.87	1.085	0.984	0.78	0.74	0.82	+1	3	0.09	11
Series-constant MAD, whole record	0.961	0.88	0.986	1.461	0.86	0.84	0.88	+1	1	0.02	11

Note: IQR is the interquartile range and SD is the standard deviation of the increments.

Online conformal recalibration as a reference. We apply a spread recalibration of the adaptive conformal inference (ACI) type, under the same information constraint as the deployed online layer, to the ACP prior, the equal-weight pool, and the GRU ensemble, which has the lowest training-window CRPS among the members (Table S18). For each lead time the target miscoverage α_t of the central interval starts at $\alpha = 0.10$ and is updated on every resolved pair as $\alpha_{t+1} = \alpha_t + \gamma_{\mathrm{ACI}}(\alpha - e_t)$, with α_t clipped to $[0.001, 0.5]$. Here $e_t = 1$ if the outcome of the pair fell outside the central interval of the base distribution at the miscoverage level in force when the pair was issued, and $e_t = 0$ otherwise. The issued distribution is the base distribution dilated about its median by the ratio, clipped to $[0.5, 3]$, of the width of its central $(1 - \alpha_t)$ interval to that of its central 90% interval. Before clipping, the issued central 90% interval has the width of the base $(1 - \alpha_t)$ interval, and its endpoints can differ from those of that interval when the base distribution is asymmetric about its median. The indicator e_t refers to the base interval, whereas ACI in its original form updates on the misses of the interval it issues. Only pairs resolved strictly before the origin enter the update. The step size γ_{ACI} is varied over $\{0.005, 0.01, 0.02\}$, and Table 8 of the main text uses the middle value, $\gamma_{\mathrm{ACI}} = 0.01$. The smallest step size gives the recalibrated prior its lowest CRPS on both projects. Over the three step sizes its Project-B CRPS lies between 1.886 and 1.923, against 1.848 for the deployed ACP, and at each step size the paired interval of the difference includes zero.

Alternative global sequence member. In Table S19, the GRU ensemble is replaced by a Transformer-encoder ensemble with the same inputs, training protocol, ensemble size, and output interface. We score this ensemble as a single member, substitute it for the GRU ensemble at the same weight in the deployed prior under the online layer, and repeat the full selection procedure and the online evaluation on the alternative member set. The alternative member embeds each of the 16 input steps linearly into 32 dimensions, adds a learned positional embedding, and applies two post-norm encoder layers with four attention heads, a

Table S18: Adaptive conformal spread recalibration as a reference method.

Configuration	γ_{ACI}	A CRPS	A C.90	B CRPS	B C.90	B C.90 H	B C.90 V	Lead	F.ep.
ACP prior (static)	–	1.032	0.97	1.879	0.93	0.93	0.94	+1	1
ACP prior + ACI	0.005	1.007	0.90	1.886	0.93	0.92	0.94	+1	1
ACP prior + ACI	0.010	1.013	0.91	1.895	0.92	0.92	0.92	+1	1
ACP prior + ACI	0.020	1.070	0.89	1.923	0.90	0.90	0.90	+1	1
Equal-weight pool (static)	–	1.122	0.86	1.534	0.81	0.82	0.80	+1	1
Equal-weight pool + ACI	0.005	1.127	0.90	1.533	0.84	0.86	0.81	+1	1
Equal-weight pool + ACI	0.010	1.129	0.88	1.536	0.85	0.87	0.82	+1	1
Equal-weight pool + ACI	0.020	1.139	0.85	1.549	0.85	0.86	0.84	+1	1
GRU ensemble (static)	–	1.066	0.86	2.706	0.68	0.78	0.57	–	1
GRU ensemble + ACI	0.005	1.046	0.90	2.655	0.75	0.82	0.66	–	1
GRU ensemble + ACI	0.010	1.048	0.88	2.656	0.78	0.85	0.69	–	1
GRU ensemble + ACI	0.020	1.060	0.87	2.669	0.76	0.84	0.68	–	1
ACP (deployed)	–	1.008	0.88	1.848	0.85	0.86	0.83	+1	2

feed-forward width of 64, ReLU activation, and no dropout. The encoder output is averaged over the observed positions and passed with the lead and family inputs to the same 34-32-1 head as the GRU member in Table S1.

Table S19: Robustness to the choice of the global sequence member.

Configuration	A CRPS	A C.90	A w.stg	B CRPS	B C.90	B C.90 H	B C.90 V	Lead	F.ep.	Burden	Rate det.
Transformer ensemble, single member	1.070	0.84	1.102	2.207	0.74	0.78	0.69	–	3	0.06	7
GRU ensemble, single member	1.066	0.86	1.097	2.706	0.68	0.78	0.57	–	1	0.06	3
ACP prior, Transformer in place of GRU + online	1.010	0.88	1.038	1.839	0.85	0.87	0.83	+1	2	0.03	12
ACP (deployed)	1.008	0.88	1.036	1.848	0.85	0.86	0.83	+1	2	0.03	11
Transformer member set, full proc. + online	1.010	0.88	1.037	1.896	0.85	0.88	0.82	+1	1	0.02	10

Selection variability across optimiser seeds. The full deployment rule is applied to the fronts obtained with 10 random seeds per family (Table S20).

Table S20: Deployment-rule selection across NSGA-II seeds.

Seed	Family	τ	LLT	GP	Hyperbola	LightGBM	GRU	Min. cov.	Test CRPS	Test C.90	w.stg
1	SLP	1.69	0.31	0.05	0.00	0.58	0.05	0.848	1.032	0.97	1.056
2	QA	1.77	0.26	0.01	0.00	0.71	0.01	0.852	1.033	0.97	1.057
3	QA	1.78	0.24	0.08	0.00	0.68	0.00	0.855	1.038	0.97	1.062
4	SLP	1.66	0.28	0.05	0.01	0.62	0.04	0.841	1.028	0.97	1.052
5	SLP	1.65	0.37	0.09	0.01	0.47	0.05	0.855	1.037	0.97	1.061
6	SLP	1.64	0.22	0.12	0.00	0.66	0.00	0.838	1.027	0.97	1.051
7	QA	1.75	0.28	0.07	0.02	0.57	0.07	0.841	1.040	0.98	1.063
8	SLP	1.67	0.35	0.03	0.00	0.53	0.09	0.838	1.031	0.97	1.055
9	QA	1.79	0.25	0.09	0.00	0.66	0.00	0.855	1.041	0.97	1.065
10	QA	1.80	0.24	0.11	0.00	0.64	0.01	0.855	1.044	0.98	1.068

Note: Test scores are those of the selected static prior before online adaptation.

S10. Crossed-group, stage-wise, and member-level diagnostics

Crossed calibration groups. Tables S21 and S22 summarise the coverage, width, and pair counts of the deployed ACP, its static prior, and the reference pools in the groups formed by

crossing lead time with instrument family on the Project-A held-out window and with response type on the Project-B replay. The 95% intervals are moving-block-bootstrap intervals over origin dates, with five-day blocks on Project A and three-day blocks over at most 17 origins per lead on Project B.

Table S21: Crossed-group coverage and width on the Project-A held-out window.

Pool	Group	n	Cov. 90 %	95 % CI	Cov. 80 %	Width_{80}	CRPS
ACP (deployed)	$h = 1$, inclinometer	180	0.92	[0.86, 0.97]	0.82	2.59	0.584
	$h = 1$, crown	600	0.87	[0.84, 0.90]	0.76	2.62	0.598
	$h = 3$, inclinometer	174	0.93	[0.88, 0.96]	0.83	3.95	0.798
	$h = 3$, crown	580	0.88	[0.83, 0.91]	0.77	4.32	1.001
	$h = 5$, inclinometer	228	0.92	[0.86, 0.96]	0.80	4.34	0.917
	$h = 5$, crown	760	0.88	[0.81, 0.94]	0.76	5.02	1.122
	$h = 7$, inclinometer	168	0.90	[0.82, 0.96]	0.83	5.62	1.165
	$h = 7$, crown	560	0.86	[0.79, 0.91]	0.76	6.52	1.494
ACP prior, static	$h = 1$, inclinometer	180	0.98	[0.96, 0.99]	0.94	3.53	0.604
	$h = 1$, crown	600	0.99	[0.98, 1.00]	0.94	3.61	0.605
	$h = 3$, inclinometer	174	0.98	[0.96, 1.00]	0.94	5.39	0.841
	$h = 3$, crown	580	0.97	[0.96, 0.99]	0.89	5.88	1.011
	$h = 5$, inclinometer	228	0.99	[0.97, 1.00]	0.94	6.11	0.964
	$h = 5$, crown	760	0.97	[0.95, 0.99]	0.91	6.85	1.150
	$h = 7$, inclinometer	168	0.98	[0.94, 1.00]	0.92	7.62	1.215
	$h = 7$, crown	560	0.95	[0.93, 0.96]	0.87	8.50	1.520
Local-only	$h = 1$, inclinometer	180	0.94	[0.90, 0.98]	0.82	4.78	1.006
	$h = 1$, crown	600	0.86	[0.81, 0.89]	0.72	3.39	0.854
	$h = 3$, inclinometer	174	0.91	[0.88, 0.94]	0.80	5.64	1.221
	$h = 3$, crown	580	0.80	[0.76, 0.84]	0.68	4.26	1.166
	$h = 5$, inclinometer	228	0.88	[0.85, 0.91]	0.74	6.52	1.548
	$h = 5$, crown	760	0.84	[0.79, 0.90]	0.73	5.21	1.316
	$h = 7$, inclinometer	168	0.92	[0.88, 0.96]	0.79	7.64	1.676
	$h = 7$, crown	560	0.80	[0.77, 0.82]	0.66	5.99	1.694
Global-only	$h = 1$, inclinometer	180	0.96	[0.94, 0.98]	0.89	3.19	0.605
	$h = 1$, crown	600	0.98	[0.97, 0.99]	0.90	3.19	0.588
	$h = 3$, inclinometer	174	0.91	[0.86, 0.94]	0.81	3.71	0.808
	$h = 3$, crown	580	0.85	[0.82, 0.88]	0.74	3.90	0.980
	$h = 5$, inclinometer	228	0.85	[0.81, 0.89]	0.74	3.62	0.927
	$h = 5$, crown	760	0.81	[0.74, 0.88]	0.71	4.05	1.120
	$h = 7$, inclinometer	168	0.79	[0.72, 0.85]	0.65	3.98	1.193
	$h = 7$, crown	560	0.71	[0.64, 0.77]	0.61	4.56	1.517
Equal-weight	$h = 1$, inclinometer	180	0.96	[0.92, 0.99]	0.87	4.14	0.796
	$h = 1$, crown	600	0.93	[0.90, 0.95]	0.81	3.31	0.699
	$h = 3$, inclinometer	174	0.94	[0.91, 0.97]	0.84	4.86	1.014
	$h = 3$, crown	580	0.83	[0.79, 0.87]	0.72	4.12	1.047
	$h = 5$, inclinometer	228	0.89	[0.87, 0.91]	0.77	5.36	1.218
	$h = 5$, crown	760	0.84	[0.78, 0.90]	0.75	4.75	1.188
	$h = 7$, inclinometer	168	0.90	[0.85, 0.95]	0.77	6.17	1.432
	$h = 7$, crown	560	0.77	[0.73, 0.81]	0.66	5.42	1.569

Note: Width_{80} is the mean scale-normalised central 80% interval width. Intervals use five-day bootstrap blocks.

Stage-wise performance. Table S23 compares the stage-wise CRPS and coverage of the deployed ACP, its static prior, the reference pools, and the five members on the Project-A training and held-out windows. Stages 3 and 4 and the first part of stage 5 lie in the training window and are in-sample for the ACP prior, so stage 5 is split into its training and held-out parts.

Member-level transfer results. Table S24 collects the member-level scores on the Project-B replay by response type and on the Project-A held-out window by instrument family.

Table S22: Crossed-group coverage and width on the Project-B replay.

Pool	Group	n	Cov. 90 %	95 % CI	Cov. 80 %	Width$_{80}$	CRPS
ACP (deployed)	h = 1, H	85	0.87	[0.80, 0.93]	0.80	4.48	0.841
	h = 1, V	75	0.80	[0.69, 0.91]	0.73	5.28	1.887
	h = 2, H	80	0.88	[0.80, 0.95]	0.78	6.43	1.282
	h = 2, V	70	0.83	[0.74, 0.91]	0.76	8.85	2.645
	h = 3, H	75	0.83	[0.75, 0.92]	0.77	9.23	1.706
	h = 3, V	65	0.88	[0.80, 0.95]	0.85	13.24	3.121
ACP prior, static	h = 1, H	85	0.94	[0.86, 1.00]	0.86	3.78	0.831
	h = 1, V	75	0.91	[0.81, 0.97]	0.83	4.70	1.848
	h = 2, H	80	0.94	[0.86, 0.99]	0.82	5.82	1.331
	h = 2, V	70	0.94	[0.86, 1.00]	0.84	8.07	2.657
	h = 3, H	75	0.89	[0.79, 0.99]	0.84	8.11	1.799
	h = 3, V	65	0.98	[0.95, 1.00]	0.92	11.89	3.216
Local-only	h = 1, H	85	0.95	[0.89, 1.00]	0.88	3.99	0.690
	h = 1, V	75	0.83	[0.73, 0.92]	0.69	3.61	1.501
	h = 2, H	80	0.89	[0.80, 0.96]	0.81	5.13	0.956
	h = 2, V	70	0.76	[0.61, 0.89]	0.64	4.97	1.944
	h = 3, H	75	0.87	[0.79, 0.93]	0.79	6.50	1.157
	h = 3, V	65	0.88	[0.77, 0.97]	0.66	6.58	2.114
Global-only	h = 1, H	85	0.73	[0.59, 0.87]	0.58	2.00	0.878
	h = 1, V	75	0.55	[0.37, 0.69]	0.44	2.11	2.713
	h = 2, H	80	0.47	[0.34, 0.60]	0.41	2.04	1.625
	h = 2, V	70	0.36	[0.24, 0.46]	0.27	2.08	4.356
	h = 3, H	75	0.40	[0.27, 0.51]	0.28	2.11	2.370
	h = 3, V	65	0.26	[0.14, 0.37]	0.23	2.10	5.595
Equal-weight	h = 1, H	85	0.94	[0.88, 1.00]	0.86	3.19	0.653
	h = 1, V	75	0.85	[0.71, 0.97]	0.75	3.01	1.726
	h = 2, H	80	0.82	[0.72, 0.93]	0.70	3.89	0.985
	h = 2, V	70	0.77	[0.59, 0.93]	0.73	3.81	2.303
	h = 3, H	75	0.69	[0.55, 0.85]	0.64	4.74	1.278
	h = 3, V	65	0.75	[0.58, 0.91]	0.69	4.79	2.604

Note: H denotes horizontal displacement and V denotes settlement. Intervals use three-day bootstrap blocks.

Table S23: Stage-wise scale-normalised CRPS and coverage on Project A.

Pool	Stage 3, training n = 3,162		Stage 4, training n = 933		Stage 5, training n = 1,139		Stage 5, held-out n = 3,016		Stage 6, held-out n = 234	
	CRPS	C.90	CRPS	C.90	CRPS	C.90	CRPS	C.90	CRPS	C.90
ACP (deployed)	1.536	0.88	1.227	0.92	1.134	0.89	1.036	0.88	0.652	0.93
ACP prior, static	1.504	0.92	1.210	0.95	1.142	0.96	1.056	0.97	0.722	1.00
Equal-weight	1.609	0.74	1.145	0.86	1.156	0.83	1.142	0.85	0.858	0.97
Local-only	1.735	0.70	1.196	0.84	1.234	0.84	1.292	0.84	1.144	0.94
Global-only	1.560	0.74	1.327	0.79	1.200	0.79	1.039	0.84	0.637	0.96
Local linear trend	1.764	0.61	1.316	0.65	1.390	0.61	1.256	0.63	0.721	0.91
Gaussian process	1.849	0.63	1.375	0.76	1.501	0.76	1.280	0.84	0.772	0.98
Rectangular hyperbola	2.783	0.60	2.138	0.67	1.879	0.77	2.487	0.70	3.041	0.64
LightGBM	1.658	0.62	1.360	0.70	1.189	0.74	1.055	0.76	0.684	0.84
GRU ensemble	1.563	0.79	1.387	0.79	1.303	0.79	1.097	0.85	0.666	0.98

Note: Stage-6 pairs are all at the five-day lead because monitoring sessions in that stage were five days apart.

S11. Ground conditions of the two projects

Table S25 summarises the strata and the soil parameters given by the detailed site investigation reports of the two sites. The Project-A investigation was made in 2017 for an earlier building scheme on the same plot, also with a three-level basement. At Project A, fill overlies Holocene alluvial–proluvial silty clay, silt, and fine sand on sandy cobble that contains lenses

Table S24: Member-level and pooled scores by response type and instrument family.

Member	B: horizontal			B: settlement			A: all			A: crown			A: inclinometer		
	CRPS	C.90	Width_{80}	CRPS	C.90	Width_{80}	CRPS	C.90	Width_{80}	CRPS	C.90	Width_{80}	CRPS	C.90	Width_{80}
ACP (deployed)	1.258	0.86	6.61	2.522	0.83	8.94	1.008	0.88	4.50	1.051	0.87	4.62	0.865	0.92	4.12
Local linear trend	1.325	0.75	5.40	2.592	0.64	5.14	1.217	0.65	3.15	1.224	0.66	3.20	1.194	0.63	3.00
Gaussian process	1.177	0.84	4.89	2.809	0.79	5.51	1.243	0.85	5.29	1.287	0.84	5.20	1.096	0.91	5.59
Rectangular hyperbola	1.224	0.87	5.16	1.915	0.76	4.31	2.527	0.70	6.73	2.314	0.67	5.79	3.238	0.80	9.86
LightGBM	2.052	0.09	0.24	4.171	0.04	0.26	1.029	0.77	3.37	1.085	0.76	3.53	0.841	0.81	2.84
GRU ensemble	1.346	0.78	3.85	4.260	0.57	3.94	1.066	0.86	4.33	1.073	0.86	4.32	1.042	0.88	4.39

of medium sand, ranges from loose to dense, and was explored to depths of 21.3 to 28.6 m. Groundwater is phreatic water in the sandy cobble, whose permeability coefficient the report estimates from local experience at about 20 m/d. During the investigation the water table lay 8.4 to 10.1 m below the ground surface, with an annual variation of 2 to 3 m, and the report recommends pipe-well dewatering for the basement. At Project B, fill overlies stiff, weakly expansive clay, with soft mucky clay between them in places, and the stiff clay rests on strongly weathered mudstone, which was explored to depths of 10.0 to 15.4 m. Groundwater is mainly discontinuous perched water in the fill and clay, 1.0 to 5.0 m below the surface during the investigation.

Table S25: Strata and recommended soil parameters of the two sites.

Project	Stratum	Thickness (m)	Unit weight (kN/m^3)	Cohesion (kPa)	Friction angle (°)	Compression modulus (MPa)
A	Fill	0.5–2.6	18.5	6	10	–
A	Silty clay	0.5–2.3	19.5	16	12	5.5
A	Silt	0.7–2.3	19.0	10	16	5.7
A	Fine sand	0.4–3.2	19.5	0	18	6.5
A	Medium sand, lenses	0.4–3.1	19.5	–	21	8.5
A	Sandy cobble, loose to dense	–	20.0–23.0	–	23–40	18.0–46.0
B	Fill	–	18.5	–	–	3.0
B	Mucky clay, local	1.8–3.5	18.0	12	11	3.0
B	Clay, stiff	–	20.0	35	14	9.0
B	Mudstone, strongly weathered	–	22.0	50	30	–

Note: Thicknesses follow the stratum descriptions, and the other values are those recommended by the reports. Dashes mark values not given.